MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6021W 1/ AND 6021WSPL 2/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification sheet and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, subminiature, medium Mu

Outline	---	3-1 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential
Base connections:		

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2a	2g	h	2k	1k	h	1g	1a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk/k	Rg/g	Ib/b	Ic/c	Pp/p	TE	Alt
Unit:	V	V dc	V dc	V	Ohms	Meg	mA dc	mA dc	W	°C	ft
Maximum:	6.6	165	0	200	---	1.1	22	5.5	0.7	220	(See note 1)
Minimum:	6.0	---	-55	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	100	0	(See note 2)	150	---	---	---	---	---	---

GENERAL:

Qualification - Required

1/ Formerly type 6021
2/ See note 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:02Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)			0.4	II	Ib	4.5	8.5	mA dc
1256	Electrode current (2) (anode)	4	Ec = -6.5 V dc; Rk = 0	0.4	II	Ib	---	100	µA dc
1256	Electrode current (3) (anode) Type 6021WSPL	3,4	Ec = -5.5 V dc; Rk = 0	0.4	II	Ib	---	50	µA dc
1266	Total grid current	4,5	Eb = 150 V dc; Rk = 300 ohms; Rg = 1.0 Meg	0.4	II	Ic	0	-0.3	µA dc
1301	Heater current			0.4	II	If	280	320	mA
1306	Transconductance (1)	4		0.4	II	Sm	4,450	6,350	µmhos
1336	Heater-cathode leakage	4		0.4	II	Ihk	---	5.0	µA dc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	4		2.5	I	---	---	---	---
1031	High-frequency vibration	4	Rp = 10,000 ohms	6.5	Code H	Ep	---	50	mV ac
1246	Audio frequency noise	6,7	Esig = 65 mV ac; Rg = 0.1 Meg; Rp = 0.01 Meg; Rk = 75 ohms	2.5	I	---	---	---	---
1256	Electrode current (1) (anode) (difference between sections)			2.5	I	Ib	---	1.6	mA dc
1266	Grid emission	4,8	Ef = 7.5 V; Ec = -7.5 V dc; Eb = 150 V dc; Rk = 0; Rg = 1.0 Meg	2.5	I	Ic	0	-0.5	µA dc
1306	Transconductance (2)	4	Ef = 5.7 V	2.5	I	$\frac{\Delta Sm}{Ef}$	---	15	%
1316	Amplification factor	4		6.5	S3	Mu	30	40	---
1331	Direct-interelectrode capacitance	4	No shield	6.5	Code E	Cgp	1.2	1.8	pF
		4	No shield			Cin	1.8	3.0	pF
			No shield; section 1			Cout	0.20	0.36	pF
			No shield; section 2			Cout	0.22	0.42	pF
			No shield			Cgg	---	0.013	pF
			No shield			Cpp	---	0.52	pF
1231	Pulse emission	4,9	Ef = 6.0 V; e pulse = 50 v; tp = 25 µs; prr = 200 pps	6.5	S3	is	300	---	ma

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:02Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
---	Pulse cathode current (1) Type 6021WSPL	3,4,10	Ef = 6.3 V; eb = 150 v; Ec = -25 V dc; egk = +30 v	2.5	I	ik Δik(tp)	330 ---	--- 10	mA %
---	Pulse cathode current (2) Type 6021WSPL	3,4,10	Ef = 5.9 V; eb = 150 v; Ec = -25 V dc; egk = +30 v	6.5	S3	ik	310	---	mA
1116	Lead fatigue			2.5	Code E	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1041	Shock	11,12	450 G; ehk = +100 V dc; Rg = 0.1 Meg	6.5	See note 13	---	---	---	---
1031	Vibration-fatigue			6.5	See note 13	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	High-frequency vibration	4	Rp = 10,000 ohms	---	---	Ep	---	200	mV a
1336	Heater-cathode leakage	4		---	---	Ihk	---	20	μA d
1306	Change in trans-conductance (1) of individual tubes	4		---	---	ΔSm t	---	20	%
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; 1 min on, 4 min off; Ehk = 140 V ac; Ec = Eb = 0	---	---	---	---
1516	Stability life		Ehk = +200 V dc; Rg/g = 1.0 Meg; TA = room	---	---	---	---
---	Stability life-test end point: (2 and 20 hours)						
1306	Change in transconductance (1) of individual tubes	4		ΔSm t	---	15	%

Source: https://assist.dla.mil -- Downloaded: 2026l0-15T13:02Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 -Continued						
1501	Intermittent life	14, 15	Ehk = +200 V dc; Rg/g = 1.0 Meg	---	---	---	---
---	Intermittent life-test end point: (1,000 hours)						
1211	Insulation of electrodes			---	---	---	---
1501	Intermittent life (high temperature)	14, 15	Ehk = +200 V dc; Rg/g = 1.0 Meg; TE = 220°C (min)	---	---	---	---
---	Intermittent life (high temperature) - test end points: (1,000 hours)						
1266	Total grid current	4	Eb = 150 V dc; Rk = 300 ohms; Rg = 1.0 Meg	Ic	0	-0.9	µA dc
1301	Heater current			If	276	328	mA
1306	Change in transconductance (1) of individual tubes	4		ΔSm t	---	25	%
1306	Transconductance (2)	4	Ef = 5.7 V	ΔSm Ef	---	20	%
1336	Heater-cathode leakage	4		Ihk	---	10	µA dc
1211	Insulation of electrodes	4		---	---	---	---

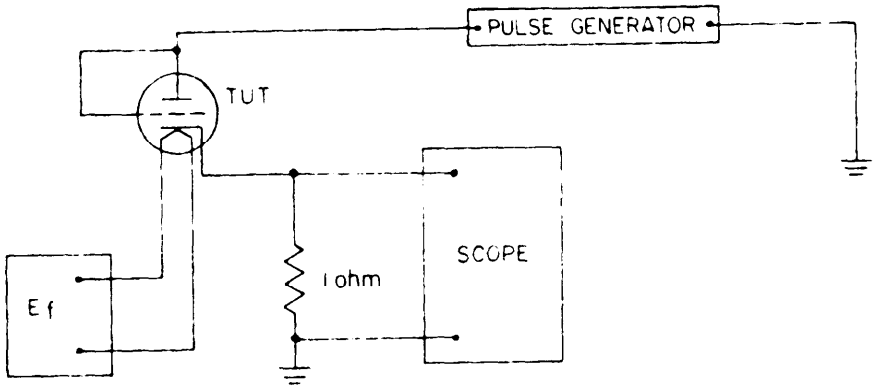
NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. The reference point for heater-cathode potential shall be the positive terminal of the cathode resistor, unless otherwise specified.
3. The additional test requirements specified for tube type 6021WSPL are intended to provide a tube that will consistantly operate satisfactorily in the PRF circuit of the modulator unit of the AN/ARN-21 Tacan set.
4. Test each unit separately.
5. This test shall be performed at the conclusion of the holding period.
6. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
7. The rejection level shall be set at the VU meter reading obtained during calibration.
8. Prior to this test, the TUT shall be preheated a minimum of 5 minutes with both sections operating at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef V	Ec V dc	Eb V dc	Rk Ohms	Rg Meg
7.5	0	150	500	1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:02Z
Check the source to verify that this is the current version before use.

9. The pulse is essentially a square wave with 1.0-µs rise time and 0.8-µs fall. The pulse shall be applied to anode and grid tied together. Pulse emission shall be measured in terms of voltage developed across a 1.0-ohm resistor in the cathode circuit. Limit shall be tested as measured by the leading edge of a calibrated oscilloscope trace, the amplitude of the trailing edge of which shall not vary by more than 20 percent from the value of the leading edge.



10. The grid pulse shall be a square wave meeting the pulse shape requirements of method 1296, and, in addition, the maximum amplitude shall occur within the first 20 percent of tp, tp = 10 µs, and prr = 1,000 pps. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to the cathode. Grid resistance, not exceeding 50 ohms, may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal.

    Peak currents shall be measured by means of a high-impedance oscilloscope, or equivalent device, connected across a cathode resistor of 1.0 ±0.01 ohm. The specified limit refers to the maximum of the pulse amplitude. The variation of the output pulse amplitude between 20 and 80 percent tp shall not exceed the specified limits for $\Delta ik(tp)$. Peak cathode current shall be read after 10 seconds or in the case of slumping peak cathode current, when a stable reading is obtained.

11. Leads may be clipped for application of voltages during impact.

12. A grid resistor of 0.1 megohm shall be added to each section, except when a thyratron-type short indicator is used.

13. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

14. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and seconds samples and shall not exceed a total of three tube failures.

15. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

16. Revision letters are not used in this revision to identify changes with respesct to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:02Z
Check the source to verify that this is the currentversion before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR
Air Force - 99
DLA - ES

User activities:
Army - ME, AV
Navy - AS, OS, MC, CG, SH
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3310)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:02Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, D. C. 20363

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20363

ATTN: ELEX 8111

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-E-11188G

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:02Z
Check the source to verify that this is the current version before use.